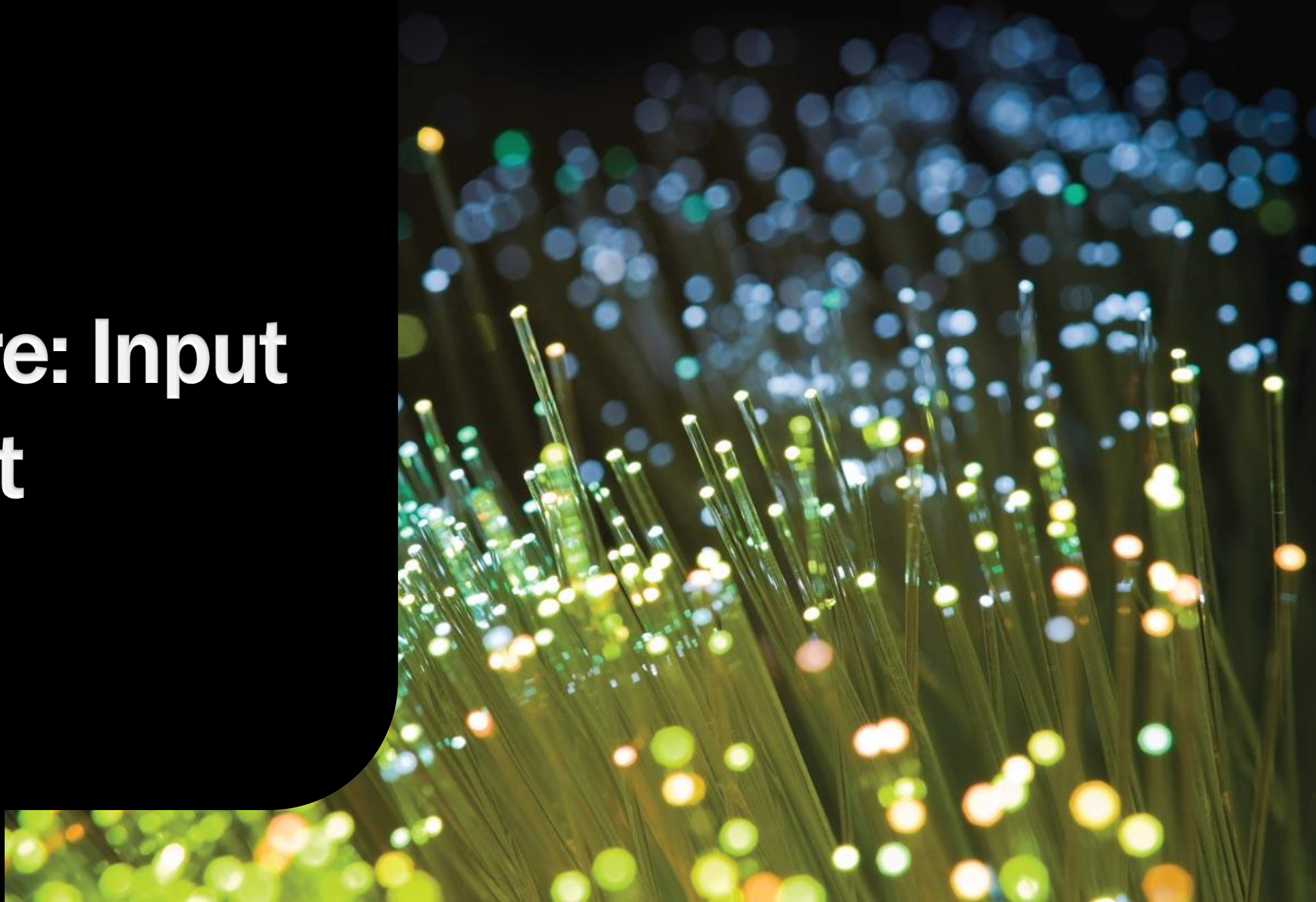


# Hardware: Input & Output

## Chapter 5



# Chapter Topics



## UNIT 5A: Input Hardware

- 5.1 Keyboards
- 5.2 Pointing Devices
- 5.3 Source Data-Entry Devices
- 5.4 The Future of Input

## UNIT 5B: Output Hardware

- 5.5 Softcopy Output: Display Screens
- 5.6 Hardcopy Output: Printers
- 5.7 Mixed Output: Sound, Voice, & Video
- 5.8 The Future of Output
- 5.9 Quality of Life: Health & Ergonomics



- **Input Hardware**

- Devices that translate data into a form the computer can process
- Translates words, numbers, sounds, and pictures into binary 0s and 1s (off or on electrical signals or light pulses)

- **Output Hardware**

- Devices that translate information processed by the computer into a form humans can understand
- Translates binary code into words, numbers, sounds, and pictures





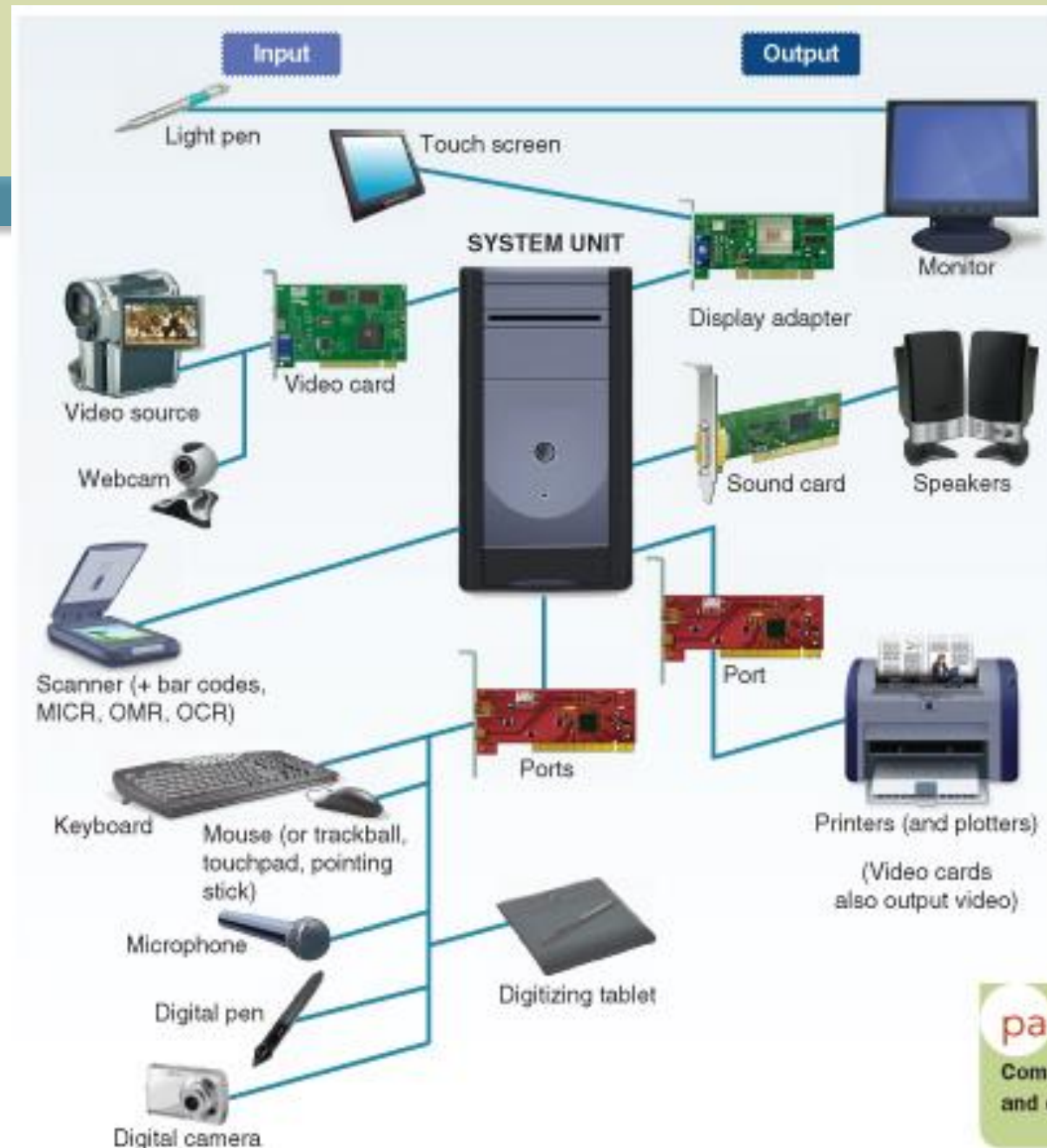


# UNIT 5A: Input Hardware

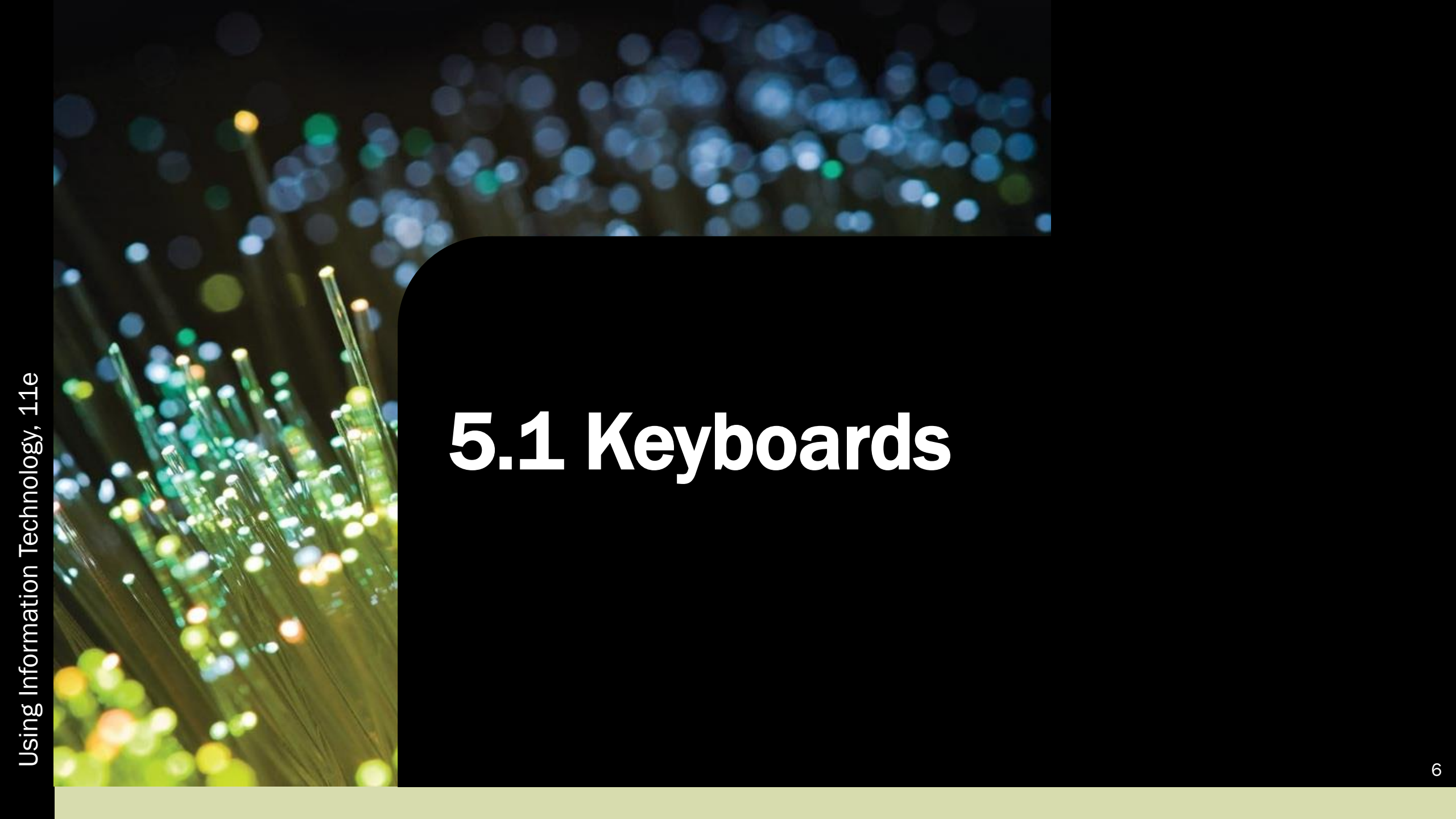


- The three major types of input hardware are **keyboards**, **pointing devices**, and **source data-entry devices**.





**panel 5.1**  
Common input  
and output devices



## 5.1 Keyboards



- **Keyboards:** convert letters, numbers, and characters into electrical signals
  - English keyboards differ from foreign language keyboards
- How keyboards work:
  - You press a key
  - This interrupts the current flowing through the circuits
  - Processor determines where the break occurs
  - It compares the location of the break with the (x,y) character map for the language on the keyboard's ROM chip
  - A character is stored in keyboard memory

(continued)





- Keyboards (*continued*)
  - The character is then sent to PC as a data stream via wire or wireless connection
  - OS interprets its own operating-system-specific commands and sends the others to the application for interpretation
  - Most keyboards are QWERTY – named for the first six letters on the top left of the keyboard.
  - Keyboards are either tactile (physical) or touch screen (virtual)







## Keyboard types

- 104 – 108 keys desktop standard
- 80 – 85 keys for laptops
- Wired
  - Connect to CPU via a serial or USB port
- Wireless use either
  - IR (infrared) technology
  - Radio Frequency (RF) technology





## Terminal Types

- **Dumb Terminals**
  - a.k.a. Video Display Terminal (VDT)
  - Has display screen and keyboard
  - Can do input and output only – *no data processing*
- **Intelligent Terminals**
  - Has screen, processor, keyboard, and memory
  - Can perform some independent functions
  - Automated teller machine; point-of-sale terminal; mobile data terminal



## 5.2 Pointing Device



*Pointing devices include the mouse and its variants, the touch screen, and various forms of pen input.*

**Pointing devices** control the position of the cursor or pointer on the screen and allow the user to select options displayed on the screen.

- **Mouse** is the principal pointing device.
  - *Mechanical mouse*: a ball inside the mouse touches the desktop surface and rolls with the mouse.
  - *Optical mouse*: uses laser beams and special chips to encode data for the computer.
  - The mouse controls the **mouse pointer** on the screen – for example, an arrow, rectangle, pointing finger.
  - When the mouse pointer changes to an I-beam, that indicates that text can be entered.
  - The mouse has one to five buttons, used for various functions, such as clicking on and dragging items on the screen.





## Pointing Devices *(continued)*

- **Trackball**

- A movable ball mounted on top of a stationary device
- Good for locations where a mouse can't move around enough



- **Touchpad**

- To use: slide your finger over this small flat surface
- Click by tapping your finger on the surface
- May require more practice to use than a mouse
- Used on laptops



- **Pointing stick**

- Located between the keys on a laptop keyboard, a pointing stick is a pressure-sensitive device that allows the user to control the pointer by directing the stick with one finger.





## Pointing Devices *(continued)*

- **Touch Screens**

- A video display screen sensitized to receive input from a finger touch.
- Used in ATMs, information, kiosks, reservation kiosks, voting machines, cellphones, tablets, and e-books.

- **Multitouch Screens**

- Display screens that allow two or more fingers or other gestures such as pinching motions to be recognized as input at any one time. It allows pinching and stretching gestures on the screen to control zooming.

- **Pen input**

- Uses a pen-like stylus for input.
- Uses handwriting recognition to translate cursive writing into data (handwriting recognition).





## Pointing Devices *(continued)*

- **Light pen**
  - A light-sensitive penlike device that uses a wired connection to a computer terminal
  - Bring the pen to the desired point on the display screen and press a button to identify the screen location
  - Used by graphics artists, engineers, and in situations that require covered hands
- **Digitizer**
  - Uses an electronic pen or puck to convert drawings and photos to digital data
  - Digitizing tablets are often used in architecture





## Pointing Devices *(continued)*

- **Digital Pen**
  - Writing instrument
  - Writers can write on paper
  - A tiny camera in the pen tip captures the writing
  - A microchip in the pen converts the pen to digital ink
  - The writing is sent as an image file to the computer
  - Some versions require special paper







## **5.3 Source Data-Entry Devices**



- **Scanning & Reading Devices**—Source data-entry devices that create machine-readable data and feed it directly into the computer (no keyboard is used)

- **Scanners**



- Use light-sensing equipment to translate images of text, drawings, and photos into digital form
- Image scanners are used in electronic imaging
- Resolution refers to the image sharpness, measured in dots per inch (dpi)
- Flatbed scanners work like photocopiers – the image is placed on the glass surface, then scanned
- Other types are sheet-fed, handheld, and drum



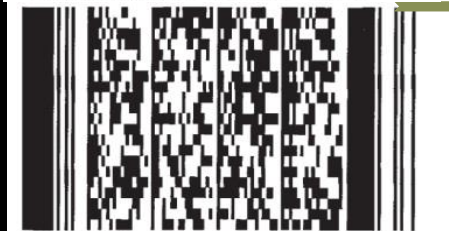
# Bar-Code Readers (source data entry)

- Photoelectric (optical) scanners that translate bar code symbols into digital code
  - The digital code is then sent to a computer
  - The computer looks up the item and displays its name and associated information

- Bar code types



Conventional 1D bar code



2D Bar code

- 1D (regular vertical stripes) holds up to 16 ASCII characters
- 2D (different-sized rectangles) can hold 1,000 to 2,000 ASCII characters
- 3D is “bumpy” code that differentiates by symbol height
  - Can be used on metal, hard rubber, other tough surfaces



## Radio-Frequency Identification (RFID)

- Based on an identifying tag bearing a microchip that contains specific code numbers. These code numbers are read by the radio waves of a scanner linked to a database.
- *Active RFID tags* have their own power source and can transmit signals over a distance to a reader device.
- *Passive RFID tags* have no battery power of their own and must be read by some sort of scanner.
- RFID tags of both types are used for a wide range of purposes and are starting to replace bar codes in many situations.







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STREET ADDRESS  
CITY, STATE

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DATE \_\_\_\_\_

PAY TO THE  
ORDER OF \_\_\_\_\_ \$ \_\_\_\_\_

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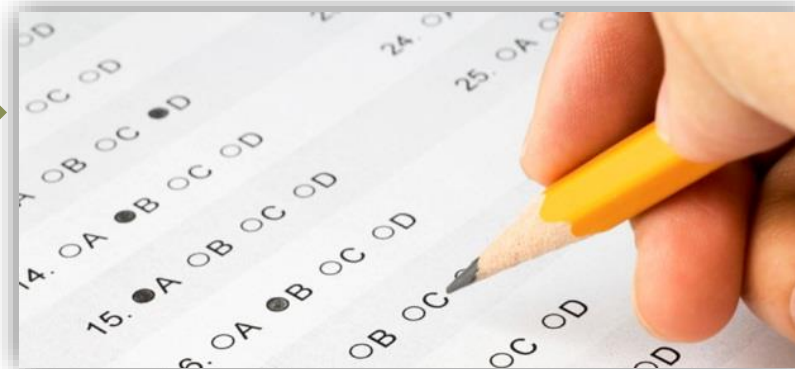
Financial Institution Name  
Street Address  
City, State

Memo \_\_\_\_\_

⑈00000101⑈ 9 ⑈0000067894⑈ 12345678⑈ ⑈0000012345⑈

## Mark Recognition Readers (source data entry)

- **MICR** – magnetic-ink character recognition
  - Uses special magnetized inks
  - Must be read by a special scanner that reads this ink
  - Used on bank checks
- **OMR** – optical mark recognition
  - Uses a special scanner that reads bubble (pencil) marks
  - Used in standardized tests such as the SAT and GRE
- **OCR** – optical character recognition
  - Converts scanned text from images (pictures of the text) to an editable text format





## Image-Capture Devices

- **Digital Cameras**

- Use a light-sensitive processor chip to capture photographic images in digital form and store them on a small disk in the camera or on flash memory cards.
  - Most can be connected to a PC by USB; smartphones include digital cameras.

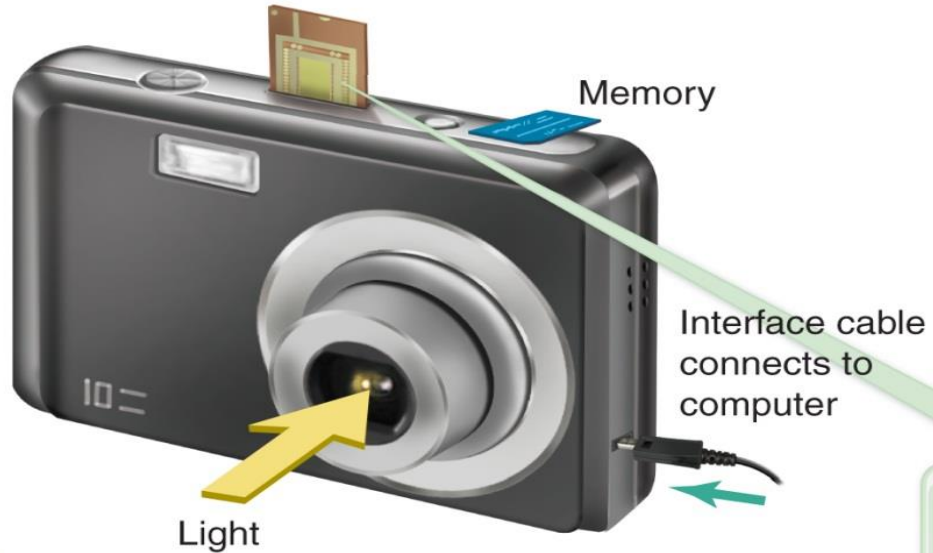
- **Webcams**

- Video cameras attached to a computer to record live moving images then post them to a website in real time.
- Can be attached externally or built into the computer/device.
- **Frame-grabber video card**
  - Can capture and digitize 1 frame at a time
- **Full-motion video card**
  - Can convert analog to digital signals at rates up to 30 frames per second
  - Looks like a motion picture





## Digital Camera



**1** Light enters the camera through the lens.

**2** The light is focused on the charge-coupled device (CCD), a solid-state chip made up of tiny, light-sensitive photosites. When light hits the CCD, it records the image electronically, just like film records images in a standard camera. The photosites convert light into electrons, which are then converted into digital information.

**3** The digital information is stored in the camera's electronic memory, either built-in or removable.

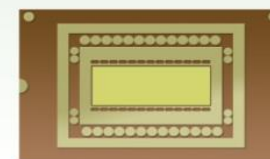
**4** Using an interface cable, the digital photo can be downloaded onto a computer, where it can be manipulated, printed, placed on a web page, or emailed.

### A look at CCDs

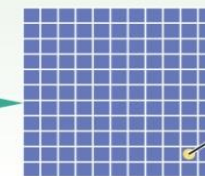
The smallest CCDs are 1/8 the size of a frame of 35mm film. The largest are the same size as a 35mm frame.



- Lower-end cameras start with 180,000 photosites.
- Professional cameras can have up to 6 million photosites.



CCD detail



Light-sensitive photosite



## Audio-Input Devices



- Record analog sound and translate it into digital files for storage and processing
- Two ways to digitize audio (often via microphone)
  - **Sound Board**
    - An add-on board in a computer that converts analog sound to digital sound, stores it, and plays it back to speakers or amp
  - **MIDI Board**
    - Stands for Musical Instrument Digital Exchange
    - Uses a standard for the interchange between musical instruments, synthesizers, and computers



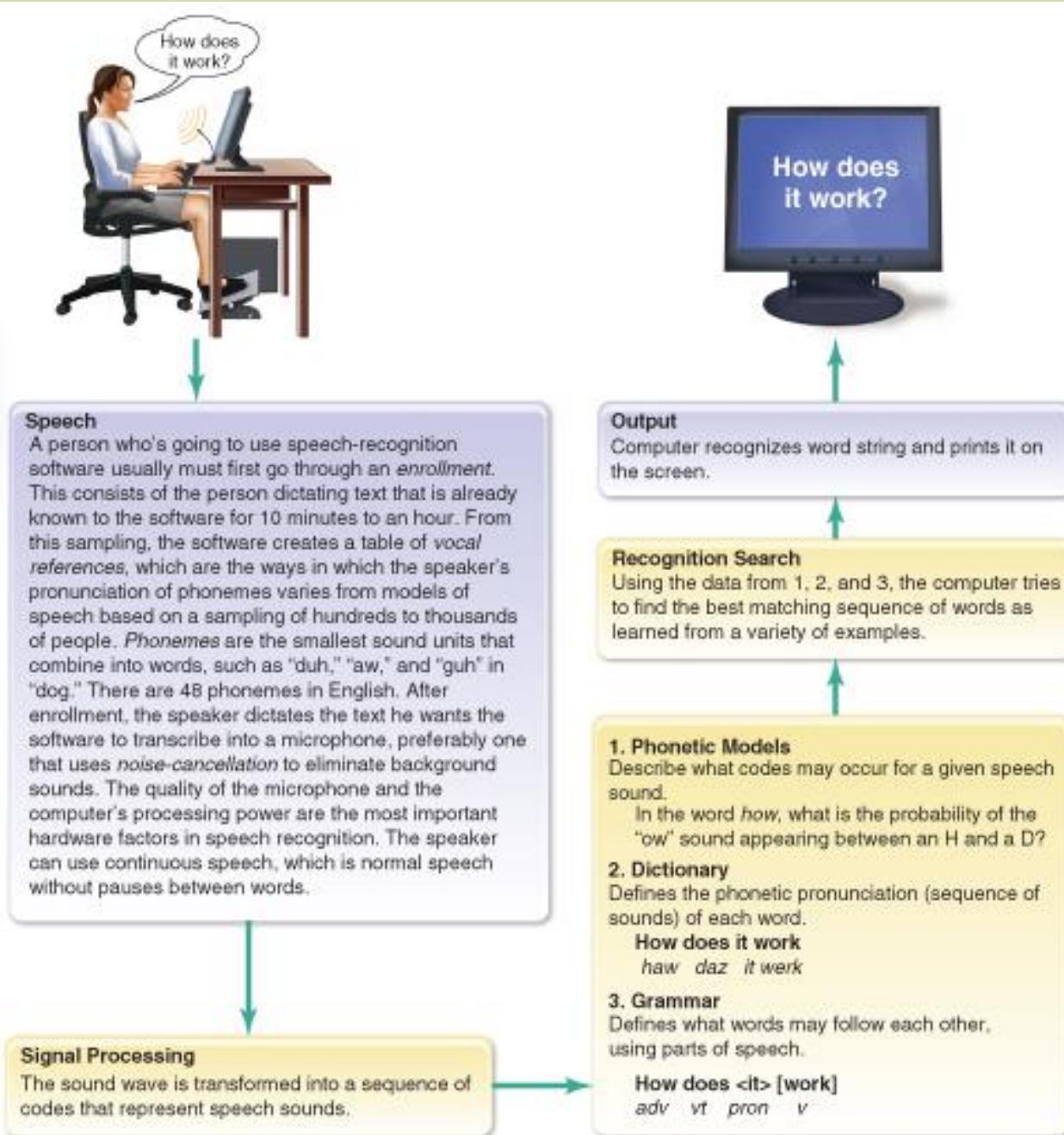
- **Speech-Recognition Systems**

- Use a microphone or telephone as an input device. Converts a person's speech into digital signals by comparing against 200,000 or so stored patterns.
  - Used in places where people need their hands free – warehouses, car radios, stock exchange trades.
  - Helpful for people with visual or physical disabilities that prevent them from using other input devices.





## Speech Recognition







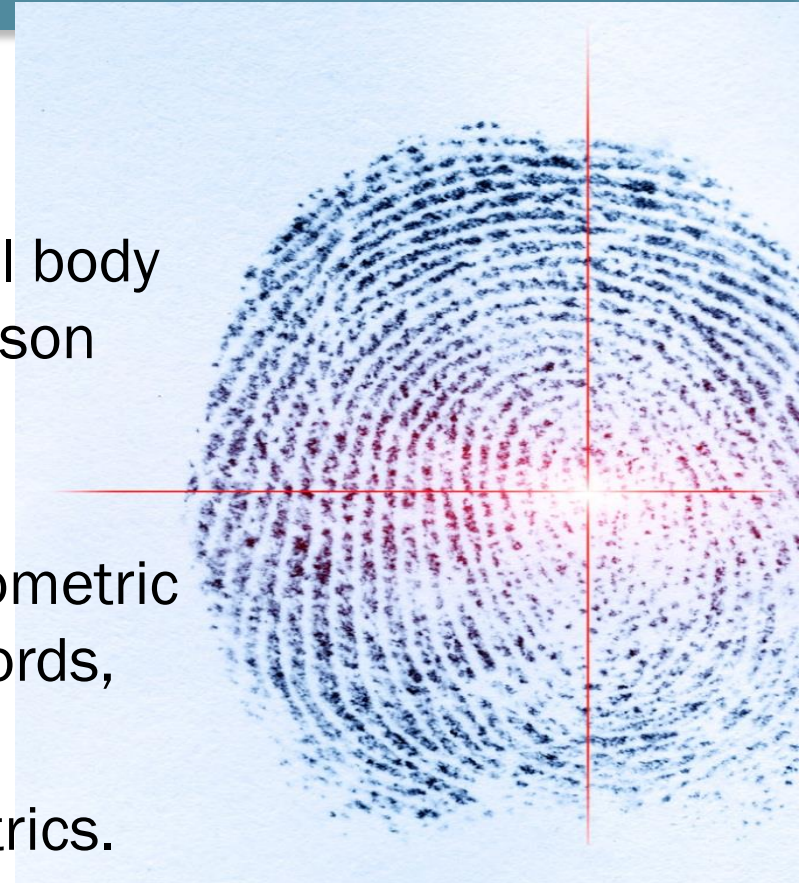
## Sensors

- Input device that collects specific data directly from the environment and transmits it to a computer.
- Can be used to detect speed, movement, weight, pressure, temperature, humidity, wind, current, fog, gas, smoke, light, shapes, images, earthquakes, etc.



## Biometric-Input Devices

- **Biometrics** is the science of measuring individual body characteristics, then using them to identify a person through a fingerprint, hand, eye, voice, or facial characteristics.
- Example: notebook computers equipped with biometric sensors that read fingerprints, instead of passwords, before allowing access to networks.
- Airport and building security systems use biometrics.





## 5.4 The Future of Input



- Data will be input from more and more locations.
- Use of source data entry will increase.
  - Better input devices for people with disabilities
  - Better speech recognition
  - Better touch and gesture-recognition input
  - Pattern recognition and improved biometrics
  - Brainwave input devices



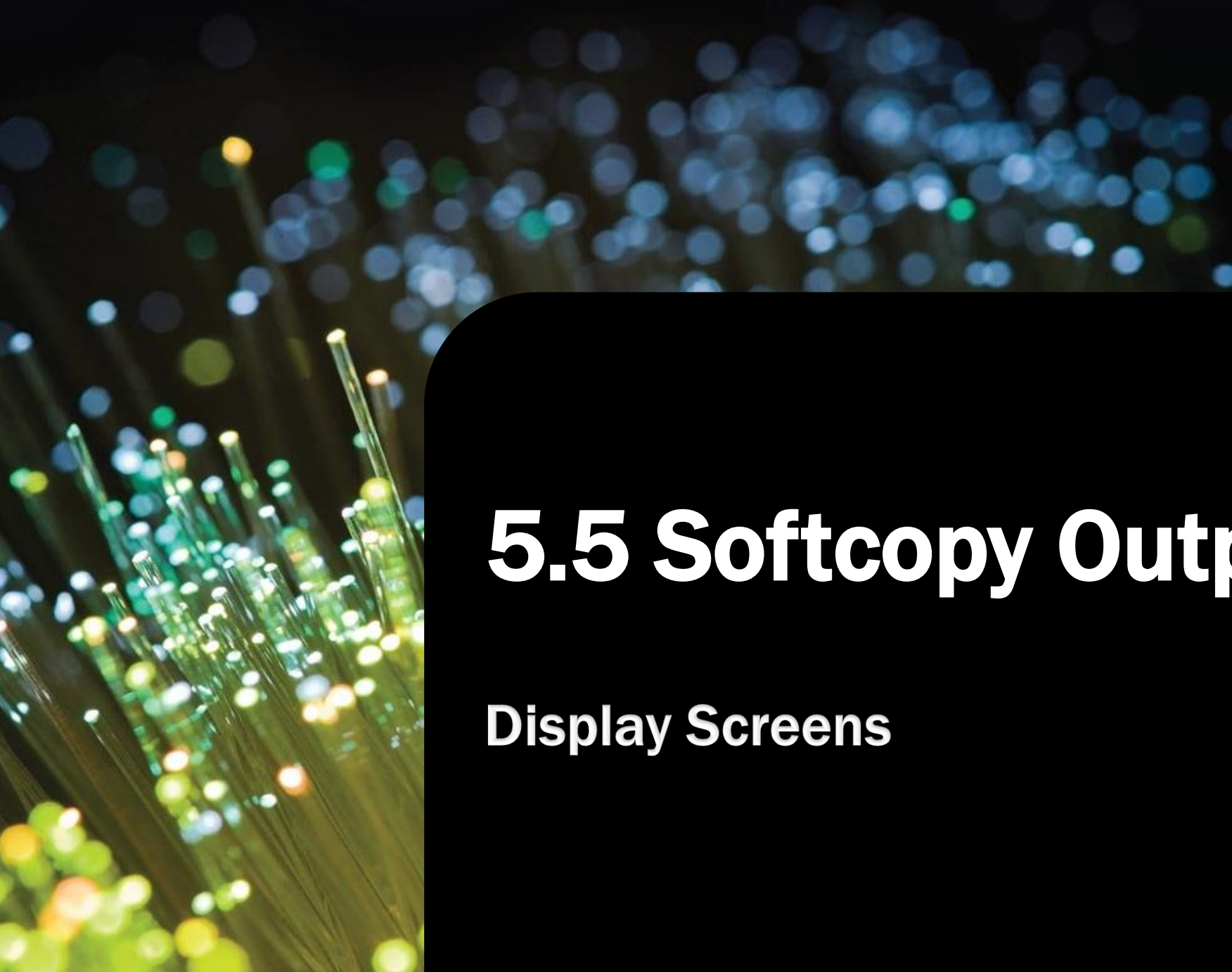




## UNIT 5B: Output Hardware



- **Softcopy** = data shown on a display screen or is in audio or voice form; it exists only electronically. This kind of output is not tangible; it cannot be touched. You can touch disks on which programs are stored, but the software itself is intangible.
- **Hardcopy** = tangible output, usually printed. The principal examples are printouts, whether text or graphics, from printers. Film, including microfilm and microfiche, is also considered hardcopy output.



# 5.5 Softcopy Output

## Display Screens



*Features of display screens to consider include screen dimension, screen clarity, and color and resolution standards.*

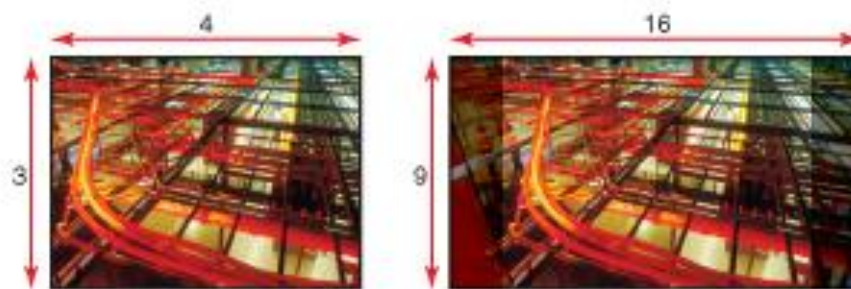
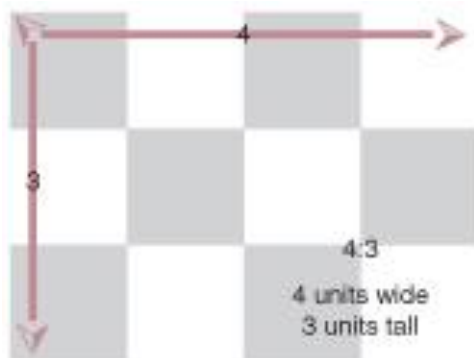
## Screen size & aspect ratio

- The **active display area** is the size of a computer screen measured diagonally from corner to corner in inches.
  - Desktop computers are commonly 15–30 inches (laptops 12–18 inches, tablets 8.4–14.1 inches, and smartphones 2.5–4.1 inches).

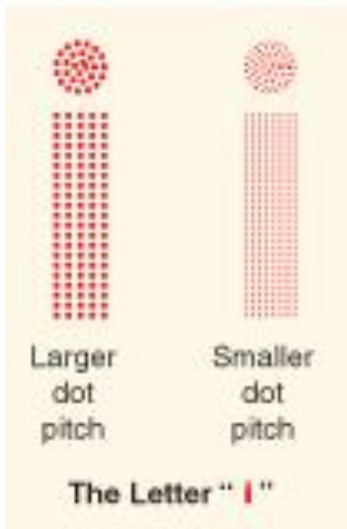




- The **aspect ratio** is the proportional relationship of a display screen's width and height.
  - Standard displays have a 4:3 aspect ratio (4 units wide to 3 units high); wide-screen displays have 16:9 or 16:10.

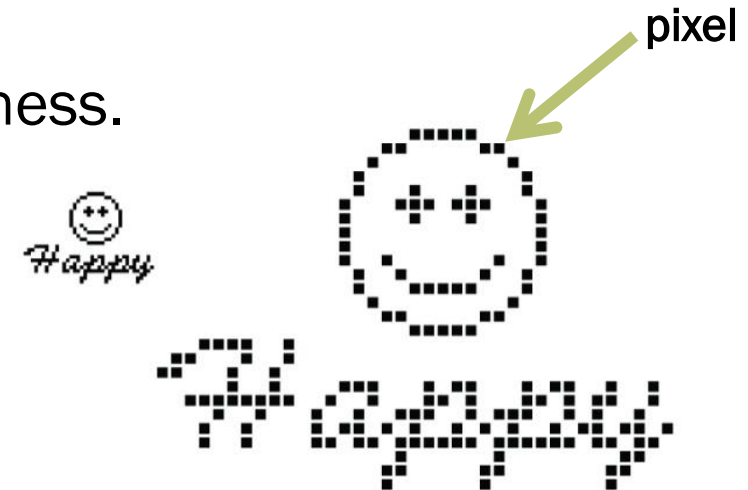






## Screen Clarity

- **Dot pitch (dp)** is the amount of space between adjacent pixels (square picture elements) on screen.
  - The closer the pixels, the crisper the image.
  - Get .25 dp or better.
- **Resolution** refers to the image sharpness.
  - The more pixels, the better the resolution.
  - Expressed in dots per inch (dpi) .





#### Standard bit depths for color

8-bit—256 colors

16-bit—65,536 colors

24-bit—16,777,216 colors

## Screen Clarity *(continued)*

- **Color depth** (bit depth) is the number of bits stored in a dot (pixel).
  - The higher the number the more true the colors.
  - 24-bit color depth is better than 8-bit color depth, but it needs more video card memory.
- **Refresh rate** is the number of times per second the pixels are recharged – a higher rate gives less flicker.



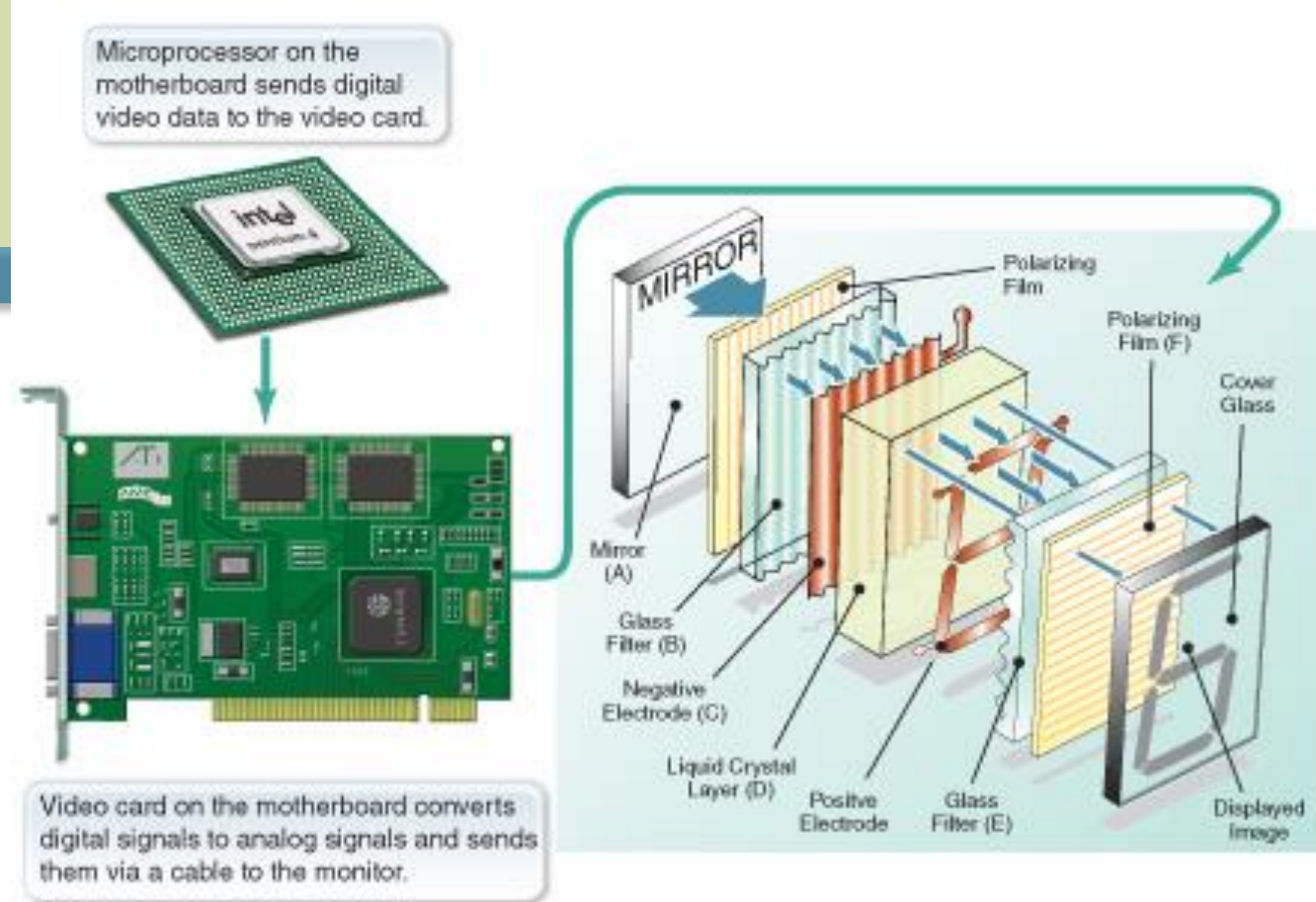
- Microcomputers come with graphics cards (video cards) that work with the screen.
- Graphics cards have their own memory (**VRAM**), which stores each pixel's information.
- The more VRAM, the higher the resolution you can use.
- Desktop publishers, graphics artists, and gamers need lots of VRAM.





## Types of Display Devices

- The most common type of display **flat-panel displays** are made of two plates of glass separated by a layer of a substance in which light is manipulated.
- One type of flat-panel display is the **liquid crystal display (LCD)**, in which molecules of liquid crystal line up in a way that alters their optical properties, creating images on the screen by transmitting or blocking light.







## Other types of displays:

- **Plasma displays:** A layer of gas is sandwiched between two glass plates, and when voltage is applied, the gas releases ultraviolet light, which activates the pixels on the screen and forms an image. Although expensive, plasma monitors offer brighter colors and screen sizes up to 150 inches wide; however, they usually do not last as long as other display types.
- Cathode-ray tube (CRT) and others: Falling out of use.
- Multiple screens: Splitting the monitor display area into multiple screens, to view different documents at once.



# 5.6 Hardcopy Output

## Printers



- Printers print text and graphics on paper or other **hardcopy** materials.
- Printer resolution is measured by **dpi (dots per inch)**; 1,200 x 1,200 is the most common for microcomputers.
- Printers are either impact or nonimpact – **impact printers** (dot-matrix printer) print by striking the paper directly; **nonimpact printers** (such as laser printers and inkjet printers) do not have direct contact with the hardcopy medium.

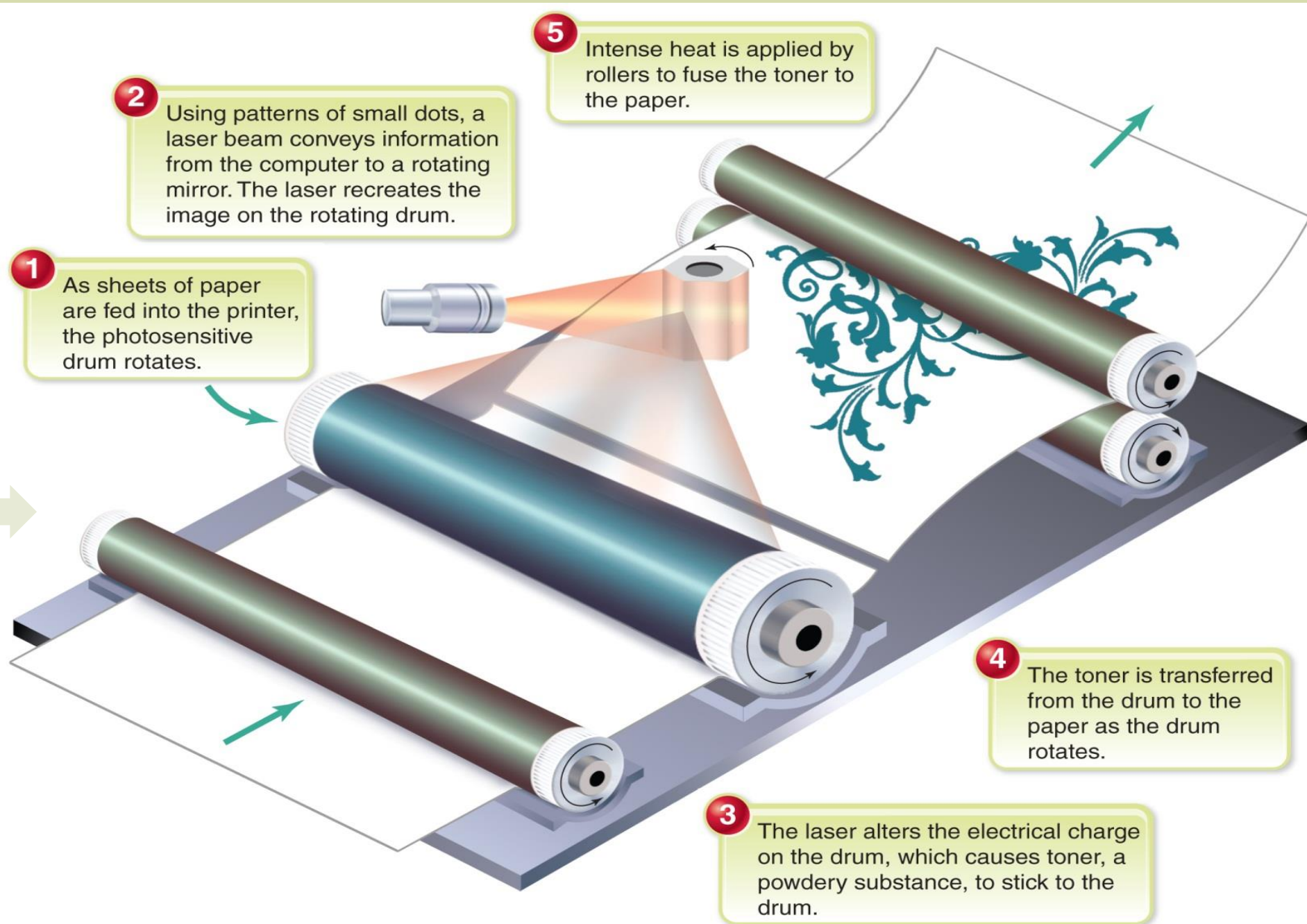




- Like a dot-matrix printer, a **laser printer** creates images with dots. However, as in a photocopying machine, these images are produced on a drum, treated with an electrically charged inklike toner (powder), and then transferred from drum to paper.
- Laser printers run with software called a **page description language (PDL)**, which tells the printer how to lay out the printed page and supports various fonts.
- A laser printer comes with one or both types of PDL: PostScript or PCL (Printer Control Language). In desktop publishing, PostScript is the preferred PDL.
- Laser printers have their own CPU, ROM, and memory (RAM), usually 16 megabytes (expandable generally up to 512 megabytes for higher-cost printers).
- When you need to print out graphics-heavy color documents, your printer will need more memory.



## Laser Printer



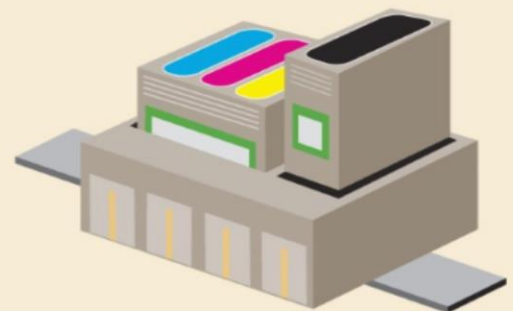


- **Inkjet printers** spray onto paper small, electrically charged droplets of ink from four or more nozzles through holes in a matrix at high speed.
- Like laser and dot-matrix printers, inkjet printers form images with little dots. Inkjet printers commonly have a dpi of 4,800 x 1,200; they spray ink onto the page a line at a time, in both high-quality black-and-white text and high-quality color graphics.

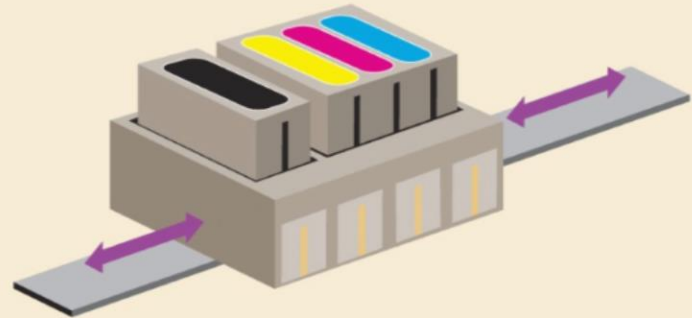


# Inkjet Printer

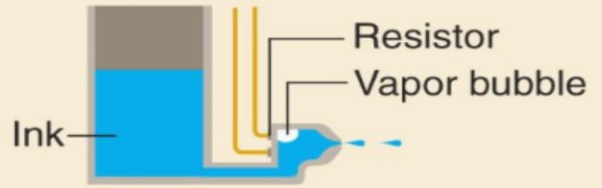
- 1 Four removable ink cartridges are attached to print heads with 64 firing chambers and nozzles apiece.



- 2 As the print heads move back and forth across the page, software instructs them where to apply dots of ink, what colors to use, and in what quantity.

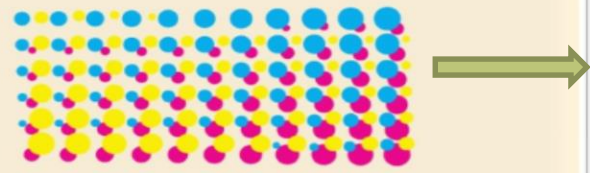


- 3 To follow those instructions, the printer sends electrical pulses to thin resistors at the base of the firing chambers behind each nozzle.



- 4 The resistor heats a thin layer of ink, which in turn forms a vapor bubble. That expansion forces ink through the nozzle and onto the paper at a rate of about 6,000 dots per second.

- 5 A matrix of dots forms characters and pictures. Colors are created by layering multiple color dots in varying densities.







## Nonimpact printers also include:

- **Thermal printers:** Low- to medium-resolution printers that use a type of coated paper that darkens when heat is applied to it; typically used in business for bar-code label applications and for printing cash register receipts.
- **Thermal wax-transfer printers:** Print a wax-based ink onto paper. After it becomes cool, the wax adheres permanently to the paper. Because of their waterfastness, these labels find uses in industrial label printing.
- **Photo printers:** Specialized machines for printing continuous-tone photo prints, with special paper and color dyes.







## Multifunction printers

- Printers that combine several capabilities:
  - Printing
  - Scanning
  - Copying
  - Faxing
- Cost less and take up less space than buying the four separate office machines.
- But if one component malfunctions, so will the other functions.





## Plotters : Designed for large-format printing

- Specialized output device designed to produce large high-quality, 3-D graphics in a variety of colors.
- Used by architects, engineers, and map-makers.
  - **Pen plotters** use one or more colored pens.
  - **Electrostatic plotters** lie partially flat on a table and use toner as photocopiers do.
  - **Large-format plotters** are large-scale inkjet printers used by graphic artists.





## **5.7 MIXED OUTPUT: SOUND, VOICE, & VIDEO**



- **Mixed Output: Sound, Voice, & Video**

- **Sound output**—produces digitized sounds, even “3-D” sound.

- You need a sound card and sound software.
    - Good speakers can improve the sound.



- **Voice output**—converts digital data into speechlike sounds.

- Used in phone trees, cars, toys and games, GPS systems, and TTS (text-to-speech) systems for hearing-impaired people.



- **Video output**—photographic images played quickly enough to appear as full-motion.

- Requires powerful processor and video card.
    - Video files are large, so a lot of storage is needed, too.
    - **Videoconferencing** is a form of video output.





## 5.8 The Future of Output



- More unusual forms of output
- More data used in (Big Data)
- More realistic output
  - Better and cheaper display screens
  - Printers that use less ink
  - Movie-quality video for PCs
  - Increased use of 3D output





# 5.9 Quality of Life

## Health & Ergonomics

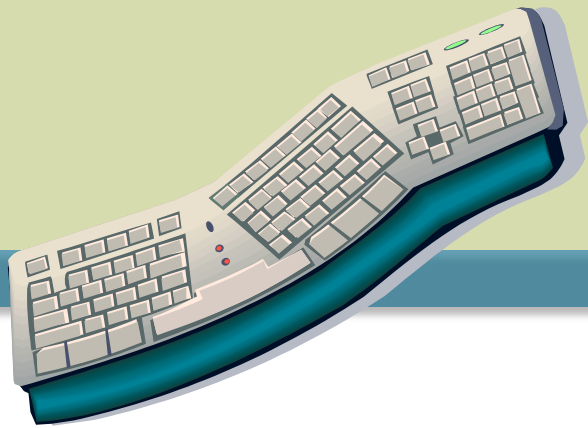


- Health Matters

- Overuse injuries and repetitive stress injuries:
  - Result when muscle groups are forced through fast, repetitive motions.
  - May effect data-entry operators who average 15,000 keystrokes an hour.
  - May effect computer users whose monitor, keyboard, and workstation are not arranged for comfort.
  - **Carpal tunnel syndrome:** Caused by pressure on the median nerve in the wrist, through short repetitive movements.
  - Eyestrain, headaches, back and neck pains can be problems.
  - Electromagnetic fields may be harmful.







**Ergonomics** is the methodology of designing a workplace to make working conditions and equipment safer and more efficient.

- Keyboards must be placed at the correct height depending on each worker's size; detachable keyboards are useful.
- Monitor refresh rates must be fast enough to avoid eyestrain.
- Monitor heights must be correct for comfortable viewing; use a tilting screen.
- Wrist rests may help avoid carpal tunnel syndrome.